

Workshop 1

Presented by:

Laura Valentina Diaz Velandia
Brandon Losada Socha
Juan Daniel Gonzalez Sierra
Frank Kenner Olmos Prada
Julian David Osorio Amaya

Teacher:

Carlos Andres Sierra Virguez

October 26, 2025



National University of Colombia
Faculty of Engineering
Department of Systems and Industrial Engineering
Software Engineering II
2025 - II

Requirements Gathering:

Functional Requirements

- FR1 The system must allow the user (student) to authenticate (login and logout).
- FR2 The system must display the complete curriculum for a selected program, organized visually by academic semesters.
- FR3 The system must allow the user to enter and save the final grade for each completed subject.
- FR4 The system must display the student's progress status (credits taken, completion percentage, etc.).
- FR5 The system must display dependencies between subjects (prerequisites).
- FR6 The system must allow filtering and searching for specific subjects by code, name, or type.
- FR7 The system must display the detailed information of each subject (credits, weekly hours, description).
- FR8 The system must provide an interface for administrators to upload and manage curriculum and course information.
- FR9 The system must generate a recommended academic path based on completed courses and their dependencies.
- FR10 The system must alert the user about potential non-compliance with prerequisites or graduation requirements.
- FR11 The system must calculate and display the semester GPA and the overall GPA based on the grades and credits of completed subjects.

Non-Functional Requirement

- NFR1 Performance: The application must load the complete curriculum in less than 3 seconds.
- NFR2 Usability: The user interface must be intuitive and easy to navigate for students with basic computer literacy.
- NFR3 Security: The system must protect user credentials (encrypted passwords) and administrator access must be strictly restricted.
- NFR4 Reliability: The system must be available 99% of the time.
- NFR5 Scalability: The design must support an increase in users without significant performance degradation (up to 5000 concurrent users).
- NFR6 Maintainability: The code must follow SOLID principles and design patterns (as discussed in the course) to facilitate future modifications.
- NFR7 Compatibility: The web interface must be fully functional on major browsers (Chrome, Firefox, Safari, Brave) and mobile devices.

User Stories

- US1: As a Student, I want to view my current progress on the curriculum visually so that I can immediately understand which subjects I have completed and what is left to do.
Acceptance Criterion: Given I am viewing the curriculum map, When I log in for the first time, Then the system displays all subjects colored by status: "Not Taken" (Grey) and "Completed" (Green).
- US2: As a Student, I want to mark a required course as 'Approved' so that my completion percentage is updated automatically and accurately.
Acceptance Criterion: Given I am on a subject details page, When I click the "Mark as Approved" button, Then the subject visual status changes to 'Approved', And my total completed credits and progress bar update instantly.
- US3: As a Student, I want to select my engineering program so that the system shows me the correct curriculum structure.
Acceptance Criterion: Given I am a new user after registration, When I select "Computing and Systems Engineering" from the program list, Then the interactive map loads all the required subjects and dependencies for that specific program.
- US4: As a Student, I want to filter the subjects by type (e.g., Mandatory, Elective) so that I can prioritize the core courses.
Acceptance Criterion: Given I am viewing the curriculum, When I apply the "Mandatory" filter, Then the map only shows the mandatory subjects, and other subjects are hidden.
- US5: As an Administrator, i want to upload a new set of curriculum data (e.g., a new semester plan) via a structured file (CSV/JSON) so that I can maintain the current academic information.
Acceptance Criterion: Given I am logged in as an administrator, When I upload a file with subject codes, names, and credits information, then the system parses the file and creates the new subject entries in the database.
- US6: As a Student, I want to see a detailed prerequisite list for any unapproved subject so that I know exactly what I need to take beforehand.
Acceptance Criterion: Given I click on an unapproved subject's card, When the subject information panel opens, Then it explicitly lists all the prerequisite subject codes and names required to enroll.
- US7: As a Student, I want to simulate taking a subject in the future so that I can plan my academic semesters effectively.
Acceptance Criterion: Given I select an unapproved subject, When I click "Simulate Enrollment," Then the system tentatively marks the subject as planned for a future semester and visually reflects the next subjects I could take.
- US8: As an Administrator, I want to edit the credit count or prerequisites for an existing subject so that curriculum changes are reflected accurately.

Acceptance Criterion: Given I search for "Software Engineering II" in the admin panel, When I change the credit count from 3 to 4 and save, Then the subject card for all students reflects 4 credits, And a log of the change is created.

- US9: As a Guest User, I want to browse the full curriculum of any program without logging in so that I can explore the programs offered in the Engineering Faculty.

Acceptance Criterion: Given I am on the main landing page, When I select a program (e.g., "Industrial Engineering"), then the full curriculum map is displayed in read-only mode without any personal progress features.

- US10: As a Student, I want the system to highlight potential prerequisites that I am missing for my next planned courses so that I avoid enrollment blockers.

Acceptance Criterion: Given I have marked some subjects as 'Planned' (US7), When I check my planned courses list, then any planned course for which I am missing a prerequisite is flagged with a warning icon.

- US11: As a Student, I want to input the final grade for each completed subject, so that my academic average is calculated automatically.

Acceptance Criterion: Given I am on the details page of a completed subject, When I enter my final grade (e.g., 4.5) and save, Then the system updates my semester GPA and my overall GPA.

